Module: Hardware Security 1
Challenge 2: Digital Filter

The device `HS102CCIT20208MBT11XQ` is a simple digital component whose top-level view is in Fig. 1. It filters two input streams entering the device via two serial single-bit transmission lines, `A` and `B`, on which data bits are transmitted synchronously with respect to a clock signal `CLK`.

In addition, the filter gets in input two control signals, `C1` and `C0` and provides, an output value `O` whose behaviour is described by the VHDL RT-level behavioural description of the listing below.

The actual implementation of `HS102CCIT20208MBT11XQ`, whose netlist is in Fig. 2, is known to contain a hardware trojan that, when activated, performs a DoS attack.



Figure 1: HS102CCIT20208MBT11XQ top-level view

```
library IEEE;
use IEEE.std_logic_1164.all;

entity HS102CCIT20208MBT11XQ is
  generic (
    PAR1 : natural := 0;
    PAR2 : natural := 7;
    PAR3 : natural := 2;
    PAR4 : natural := 1;
    PAR5 : natural := 3
    );
  port (
      clk : in std_logic;
      rst : in std_logic;
```

```
      A   : in std_logic;
      B   : in std_logic;
      C1  : in std_logic;
      C0  : in std_logic;
      O   : out std_logic);
end HS102CCIT20208MBT11XQ;

architecture beh of HS102CCIT20208MBT11XQ is

  signal A_reg  : std_logic_vector (7 downto 0);
  signal B_reg  : std_logic_vector (7 downto 0);
  signal O_reg  : std_logic_vector (7 downto 0);
  signal C0_reg : std_logic_vector (7 downto 0);
  signal C1_reg : std_logic_vector (7 downto 0);

begin

  STATE_REG_PROC: process(clk, rst)
  begin
    if(rst = '1') then
      A_reg <= (others=>'0');
      B_reg <= (others=>'0');
      O_reg <= (others=>'0');
      C0_reg <= (others=>'0');
      C1_reg <= (others=>'0');

    elsif(rising_edge(clk)) then
      A_reg <= A & A_reg(7 downto 1);
      B_reg <= B & B_reg(7 downto 1);
      C0_reg <= C0 & C0_reg(7 downto 1);
      C1_reg <= C1 & C1_reg(7 downto 1);

      case (C1 & C0) is
        when "00" =>
          O_reg <= (A_reg(PAR1) + B_reg(PAR2)) & O_reg(7 downto 1);
        when "01" =>
          O_reg <= (A_reg(PAR3) - not(B_reg(PAR4))) & O_reg(7 downto 1);
        when "10" =>
          O_reg <= O_reg(PAR5) & O_reg(7 downto 1);
      end case;

    end if;
  end process;

  OUTPUT_PROC: process(O_reg)
  begin
    O <= O_reg(7);
  end process;

end beh;
```

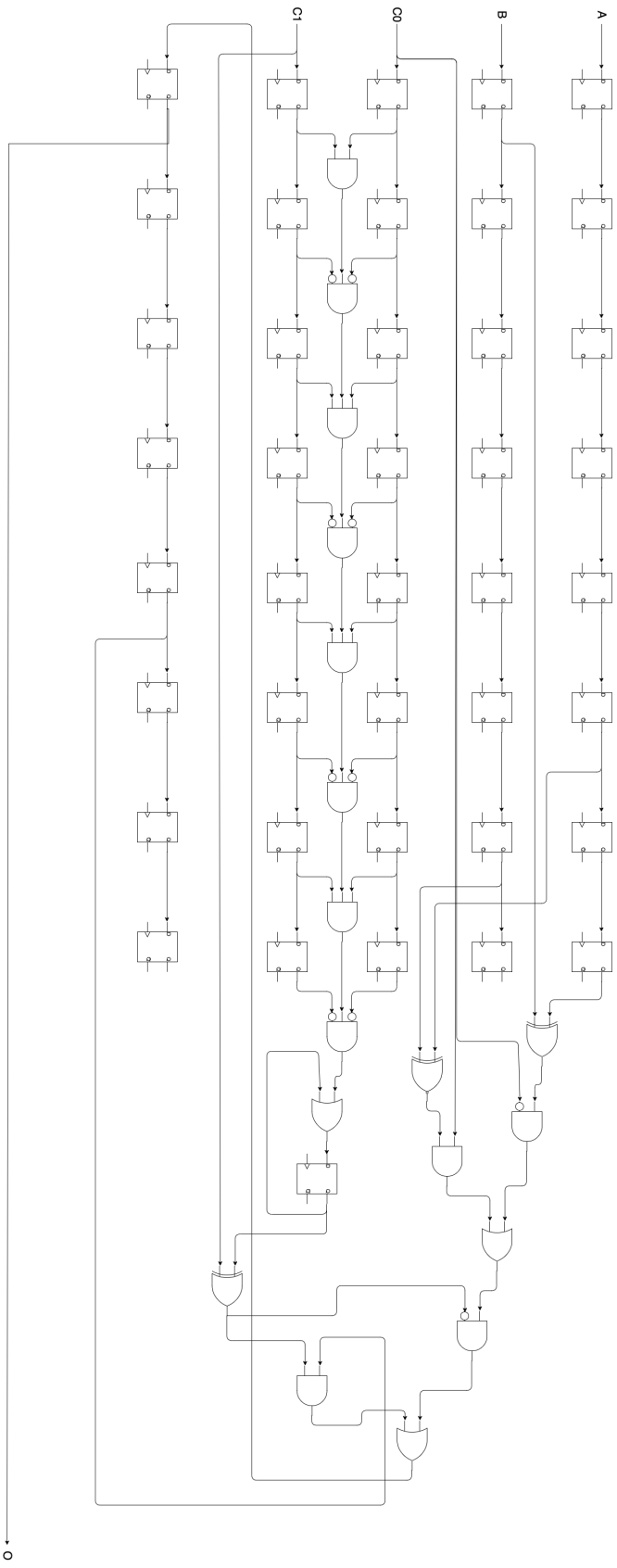


Figure 2: HS102CCIT20208MBT11XQ Netlist. CLK and RST trees are not represented, but it is implicit that these signals reach every flip-flop.